Universität Marburg

Fachbereich Mathematik und Informatik
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Forecasting: Theory and Practice

(Winter term 2026/27)

Problem Set 1

To be handed in on: Wednesday, 28.10.2026, 12:00 at Ilias,
Meeting on: Wednesday, 28.10.2026, 14:15, SR II,
Feedback handed in on: Tuesday, 03.11.2026, 23:59 at Ilias.

For the weekly submission:

- You must submit the exercise sheets in the **same group of three**.
- Only **one** group member should upload the submission and add the other group members under **"Team erstellen"**.
- The source file of the programming exercises (including code, output, and written explanations) should be compiled and submitted as a **PDF** (e.g. R Markdown or Jupyter Notebook).
- Upload your submission in exactly **one PDF** document.
- Please ensure that the submission does not contain any group members' names.

Problem 1 (*The two-piece normal distribution*)

Consider the Bank of England's use of the two-piece normal distribution for its inflation projections with density

$$f(y) = \begin{cases} (\pi/2)^{-1/2}\,(\sigma_1 + \sigma_2)^{-1} \exp\{-(y-\mu)^2/(2\sigma_1^2)\}, & y \leq \mu \\ (\pi/2)^{-1/2}\,(\sigma_1 + \sigma_2)^{-1} \exp\{-(y-\mu)^2/(2\sigma_2^2)\}, & y \geq \mu, \end{cases}$$

where $\mu \in \mathbb{R}$ and $\sigma_1, \sigma_2 > 0$.

a) Show that this is a properly normalized probability density function.

b) Calculate the expectation (mean) and variance of this distribution.

c) For right-skewed distributions, mean and median are typically greater than the mode. Show that this holds for the two-piece normal distribution if $\sigma_2 > \sigma_1$.

Problem 2 (*Bank of England fan charts*)

Consider the Bank of England CPI (consumer price index) inflation rate forecasts[1], which are given by two-piece normal distributions. The Bank of England does not specify the parameters directly in its reports, but through the mode μ, mean m and variance σ^2 of the predictive distribution. The following table describes the forecasts given in May, August, November 2021 and February, May 2022 for the inflation in the second quarter[2] of 2022. The observed inflation rate was 9.2%.

	2021			2022	
	May	Aug.	Nov.	Feb.	May
μ	2.31	3.78	4.80	7.03	9.13
m	2.31	3.78	4.80	7.05	9.13
σ^2	1.68	1.55	1.35	1.01	0.69

a) Use part b) of the previous problem to express σ_1 and σ_2 in terms of μ, m and σ^2.

b) Write a function that returns the density of the two-piece normal distribution based on μ, m and σ^2 in R.

c) Using this function, plot the densities of the five forecasts together with the corresponding true observed inflation rate (e.g. indicated by a vertical line).

d) Investigate what the German Bundesbank reports about the uncertainty of its own inflation projections. Does it follow a similar approach, and if not, how does it differ from the Bank of England's fan charts?

Problem 3 (*Forecasting inflation: LLM vs. Survey of Professional Forecasters*)

Ask an LLM of your choice to provide a probabilistic forecast of the inflation rate for the first quarter of 2027 and ask it to explain how this forecast was obtained. Next locate the corresponding forecast from the Survey of Professional Forecasters (SPF) published by the ECB and compare the two forecasts.

- How do the two forecasts differ in terms of both the predictive distribution and the reasoning given for it?
- Which of the two forecasts do you find more trustworthy and why?

Carry out the analogous exercise for a probabilistic weather forecast for Marburg: ask an LLM for a probabilistic temperature or precipitation forecast for a specific day and compare it against a forecast from a national weather service (e.g. DWD).

- How do the two forecasts differ in terms of both the predictive distribution and the reasoning given for it?
- Which of the two forecasts do you find more trustworthy and why?

[1] `https://www.bankofengland.co.uk/monetary-policy-report/monetary-policy-report`

[2] The second quarter ranges from April to June.

Problem 4 (*Post-processing ensemble forecasts with EMOS*)

Consider the provided ensemble forecasts for Frankfurt Airport, in Ilias stored in `T2M_frankfurt_airport.csv`. The variable of interest is the 2-metre temperature (T2M). The dataset covers the year 2016; for each forecast case, 50 ensemble members and the corresponding observation are available.

a) Implement a Gaussian EMOS model of the form

$$Y \mid X_1, \ldots, X_M \sim \mathcal{N}\big(a + b\,\bar{X},\ c + d\,S^2\big)\,,$$

where $\bar{X}$ and S^2 denote the ensemble mean and variance. Use January to September as training data and estimate the parameters a, b, c, d by maximum likelihood.

b) Interpret the estimated values. What do they imply about the bias and dispersion of the raw ensemble?

c) Apply the fitted model to the held-out test period (October to December). Compute a 90% prediction interval for each test observation and report the empirical coverage rate. Compare to the coverage of the raw ensemble's empirical 90% interval (5th to 95th percentile of the 50 members) and compare the mean widths of the two intervals.